

Analysis and Extension of “Disorder in Photonic Structures Induced by Random Layer Thickness”

1 Introduction

This essay focuses on the study of one-dimensional (1D) photonic crystals. As a starting point, it takes inspiration from the paper “*Disorder in Photonic Structures Induced by Random Layer Thickness*” by Chiasera et al. (2015) [Ref. 1], which investigates the effects of structural disorder due to random variations in layer thicknesses.

Based on this work, several numerical simulations were carried out to explore additional effects, such as the influence of the angle of incidence, the presence of absorption (modeled using a complex refractive index), and the use of different materials.

Separately from the study of disorder, the final part of the essay focuses on the creation of a microcavity inside a photonic crystal. This is achieved by introducing a defect in the multilayer structure, either by inserting a spacer layer made of the same material or by using a layer with a different refractive index. These configurations allow us to investigate localized modes and transmission peaks associated with optical resonances.

2 Brief analysis of the Paper

2.1 Introduction

In the paper titled “*Disorder in Photonic Structures Induced by Random Layer Thickness*”, Chiasera et al. (2015) explore the influence of structural disorder on one-dimensional photonic crystals (1DPCs) composed of alternating dielectric layers. The study focuses on the optical behavior resulting from introducing a random deviation in the thickness of the layers, and how this impacts the photonic bandgap and light transmission properties. Their work combines theoretical modeling using the transfer matrix method and experimental validation with fabricated multilayer structures. This analysis is crucial for the design of optical filters, mirrors, and sensors that rely on precise control of photonic bandgaps.

2.2 Theoretical and experimental method

2.2.1 Transfer Matrix Method

The Transfer Matrix Method (TMM) is used to model the propagation of electromagnetic waves through stratified media. In a one-dimensional photonic crystal (composed of a glass/alternating layers/air system), each layer j is characterized by a refractive index n_j and a thickness d_j . The glass has a refractive index $n_s = 1.6$, and air is assumed to have

$n_0 = 1$. The materials are considered non-magnetic and isotropic. The characteristic matrix M_j for normal incidence is given by:

$$M_j = \begin{bmatrix} \cos(\phi_j) & -\frac{i}{p_j} \sin(\phi_j) \\ -ip_j \sin(\phi_j) & \cos(\phi_j) \end{bmatrix} \quad (1)$$

where $\phi_j = \frac{2\pi}{\lambda} n_j d_j$ and $p_j = \sqrt{\frac{\epsilon_j}{\mu_j}}$ in the TE wave (and $q_j = \frac{1}{p_j}$ in the TM wave), and λ is the wavelength in vacuum. The global transfer matrix is obtained by multiplying the individual matrices of each layer:

$$M = \prod_{i=1}^N M_i$$

$$\begin{bmatrix} E_0 \\ H_0 \end{bmatrix} = \prod_{j=1}^n M_j \begin{bmatrix} E_m \\ H_m \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} E_m \\ H_m \end{bmatrix} \quad (2)$$

Denoting by E_m and H_m the electric and magnetic fields in the glass substrate, and by E_0 and H_0 the corresponding fields in air, the transfer matrix formalism allows us to relate the fields at the input and output of the multilayer structure.

The transmission coefficient t is then derived from the elements of the total matrix M :

$$t = \frac{2p_s}{(m_{11} + m_{12}p_0)p_s + (m_{21} + m_{22}p_0)} \quad (3)$$

Then the light transmission:

$$T = \frac{p_0}{p_s} |t|^2 \quad (4)$$

Analogous for the reflectivity:

$$r = \frac{(m_{11} + m_{12}p_0)p_s - (m_{21} + m_{22}p_0)}{(m_{11} + m_{12}p_0)p_s + (m_{21} + m_{22}p_0)} \quad (5)$$

And the light reflectivity:

$$R = |r|^2 \quad (6)$$

2.2.2 Refractive Index Formula

The wavelength dependence of the refractive index, known as dispersion, plays a crucial role in the optical behavior of dielectric multilayers. A common empirical model used to describe the dispersion of transparent materials is the Sellmeier equation, which is given by **[RIF 2]**:

$$n^2(\lambda) = 1 + \sum_{i=1}^3 \frac{B_i \lambda^2}{\lambda^2 - C_i} \quad (7)$$

where λ is the wavelength (typically in micrometers), and B_i and C_i are material-specific coefficients obtained from experimental data. This equation effectively models the variation of the refractive index in non-absorbing regions of the spectrum.

The dispersion behavior of the materials used in this study—silicon dioxide (SiO_2) and titanium dioxide (TiO_2)—can be described using either the Sellmeier equation or modified

empirical models that include absorption effects. For SiO₂, a transparent dielectric, the standard Sellmeier model is sufficient and typically involves three terms [Ref. 2]:

$$n_{\text{SiO}_2}^2(\lambda) = 1 + \frac{0.6961663\lambda^2}{\lambda^2 - 0.06840432} + \frac{0.4079426\lambda^2}{\lambda^2 - 0.11624142} + \frac{0.8974794\lambda^2}{\lambda^2 - 9.8961612} \quad (8)$$

This expression accurately reproduces the dispersion characteristics of SiO₂ in the range from the ultraviolet to the near-infrared. Specifically, its refractive index decreases from approximately 1.55 at $\lambda = 0.210 \mu\text{m}$ to about 1.44 at $\lambda = 1.2 \mu\text{m}$.

In contrast, TiO₂ exhibits stronger dispersion, its refractive index decreases from about 2.80 at $\lambda = 0.300 \mu\text{m}$ to roughly 2.06 at $\lambda = 1.2 \mu\text{m}$ [Ref. 2]:

$$n_{\text{TiO}_2}(\lambda) = \sqrt{4.99 + \frac{1}{96.6 \cdot \lambda^{1.1}} + \frac{1}{4.6 \cdot \lambda^{1.95}}} \quad (9)$$

Accurate knowledge of the refractive index as a function of wavelength is fundamental for designing photonic structures, as it directly affects the position and width of photonic bandgaps as well as the transmission spectra.

2.3 Sample Fabrication and Optical Characterization

The multilayer photonic crystal structures based on alternating SiO₂ and TiO₂ layers were fabricated using a multi-target radio frequency (RF) sputtering technique. To ensure layer thickness control and reproducibility, dual quartz crystal microbalances were used to monitor the deposition rates in real time. These sensors were calibrated through long (24-hour) depositions and cross-verified using m-line spectroscopy and scanning electron microscopy (SEM). The m-line technique is a waveguide-based optical method that allows precise measurement of refractive indices and film thicknesses by analyzing guided mode angles.

Optical transmittance in the visible-to-near-infrared (VIS-NIR) range was measured using a Varian-Cary double-beam spectrophotometer. This instrument compares the light transmitted through the sample with a reference beam, providing high-precision transmission spectra with a wavelength resolution of 2 nm.

2.3 Results and Discussion

The comparison between experimental and simulated transmittance spectra confirms the reliability of the Transfer Matrix Method (TMM) when the refractive index dispersion of SiO₂ and TiO₂ is properly taken into account. For the **ordered photonic crystal**, each layer was considered to have a thickness of 100 nm, for a total of 14.5 bilayers. The simulation, shown as a red curve, closely matches the experimental data (blue dotted curve), demonstrating that the model accurately reproduces the optical behavior of the multilayer structure.

For the **disordered photonic crystal**, a sequence of SiO₂/TiO₂ layers was fabricated with randomly varying thicknesses according to the rule $d = 80 + n$ nm, with $0 \leq n \leq 40$. This creates a random distribution of thicknesses between 80 nm and 120 nm. The corresponding experimental transmission spectrum (blue dotted curve) and the simulated one (red curve), based on the same random thicknesses and dispersion data, show very good agreement.

One of the main effects observed is the **broadening of the stop band**. While the ordered structure shows a clear photonic bandgap centered around 750 nm, in the disordered case the stop band becomes wider, extending from approximately 550 nm to 750 nm. Only two narrow and weak transmission peaks remain visible around 575 nm and 645 nm.

These results clearly show how structural disorder significantly affects the optical response of the photonic crystal. In particular, the introduction of randomness leads to a wider stop band and altered transmission characteristics across the visible range.

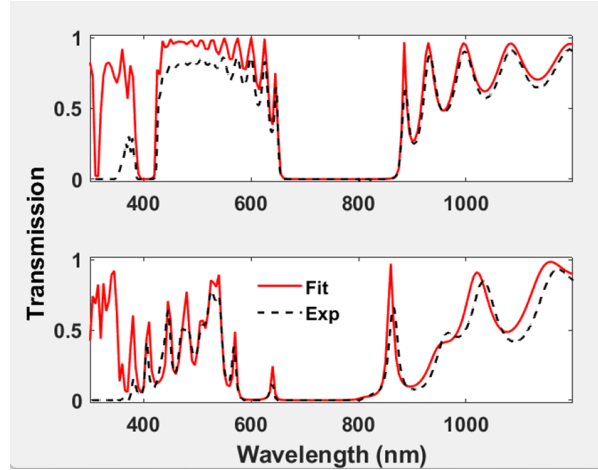


Figure 1: Experimental data and simulation of TiO₂/SiO₂

Comment on the 300–400 nm Region

A noticeable discrepancy between the simulated and experimental transmission spectra is observed in the wavelength range between 300 and 400 nm. In this region, the simulation tends to overestimate the transmittance compared to the experimental data.

This difference is mainly due to the fact that the simulation does not include the **imaginary part of the refractive index**, which accounts for *material absorption*. In particular, both SiO₂ and TiO₂ exhibit non-negligible absorption in the near-ultraviolet region, which leads to a reduction in transmitted light that is observed experimentally but not reproduced in the model.

To improve the accuracy of the simulation in this spectral range, it would be necessary to introduce a *complex refractive index*.

3 Inclusion of Absorption via Complex Refractive Index

To better reproduce the experimental results, especially in the ultraviolet range, we extended the refractive index model to include absorption by considering a **complex refractive index** of the form:

$$\tilde{n}(\lambda) = n(\lambda) + i k(\lambda) \quad (10)$$

where $n(\lambda)$ represents the real part (dispersion) and $k(\lambda)$ the imaginary part (absorption) of the refractive index, both wavelength-dependent.

The values of $k(\lambda)$ were obtained from experimental data available in literature: for SiO_2 , the absorption data was taken from [Ref. 3], and for TiO_2 from [Ref. 4]. These datasets were interpolated over the wavelength range from 300 nm to 1200 nm, with a step size of 5 nm, in order to be integrated consistently into the simulation framework.

By incorporating this imaginary part into the transfer matrix method, the model was able to better capture the reduced transmittance observed experimentally, especially in the 300–400 nm region where absorption is non-negligible.

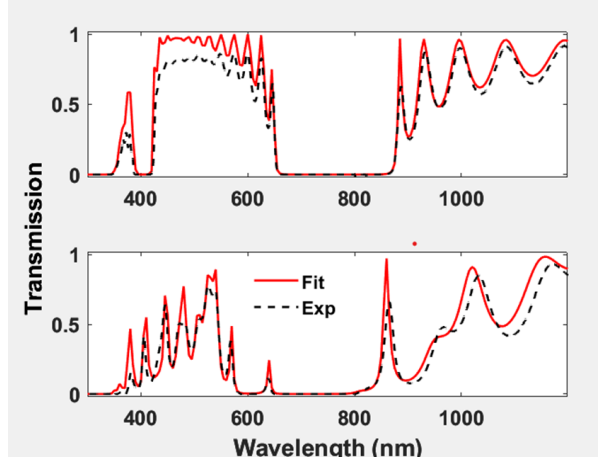


Figure 2: Experimental data and simulation of $\text{TiO}_2/\text{SiO}_2$ with absorption

4 Angular Dependence

When light impinges on a multilayer structure at an angle, the propagation within each layer is affected by the angle of incidence. This leads to a change in the phase acquired by the wave and, consequently, alters the transmission spectrum. To account for this, the transfer matrix for the k -th layer must include the angular dependence as follows [Ref. 5]:

$$M_k = \begin{bmatrix} \cos \varphi_k & -\frac{i}{p_k} \sin \varphi_k \\ -ip_k \sin \varphi_k & \cos \varphi_k \end{bmatrix} \quad (11)$$

with

$$\varphi_k = \frac{2\pi}{\lambda} n_k d_k \cos \theta_k \quad (12)$$

where n_k is the refractive index of the k -th layer, d_k its thickness, and θ_k the angle of light propagation inside the layer, related to the angle of incidence θ_0 in air by:

$$\cos \theta_k = \sqrt{1 - \left(\frac{n_0}{n_k} \sin \theta_0 \right)^2} \quad (13)$$

Additionally, we distinguish between s -polarization (TE) and p -polarization (TM) waves:

- For s -polarization: $p_k = n_k \cos \theta_k$
- For p -polarization: $p_k = \frac{\cos \theta_k}{n_k}$

The overall transfer matrix of the multilayer system, composed of s layers, is obtained by matrix multiplication as in (2). The transmission coefficient t is then calculated with (3), with p_0 and p_s also depending on the polarization:

$$\text{For } s\text{-polarization: } p_0 = n_0 \cos \theta_0, \quad p_s = n_s \sqrt{1 - \left(\frac{n_0}{n_s} \sin \theta_0 \right)^2}$$

$$\text{For } p\text{-polarization: } p_0 = \frac{\cos \theta_0}{n_0}, \quad p_s = \frac{1}{n_s} \sqrt{1 - \left(\frac{n_0}{n_s} \sin \theta_0 \right)^2}$$

Finally, the transmission of the multilayer structure is computed as (4).

This formalism allows us to analyze how the angle of incidence modifies the photonic band structure and the transmittance properties.

In general, as the angle of incidence increases, the transmittance spectrum experiences a shift towards shorter wavelengths, commonly referred to as a *blue shift*. This effect shifts (in this case) the photonic band gap by approximately 100 nm, with observed variations ranging from 75 nm to 150 nm depending on the specific configuration and wavelength.

A similar trend is observed for both TE and TM polarization modes, although slight differences may arise due to the distinct interaction of each mode with the layered medium. Nonetheless, the overall behavior remains consistent between the two polarizations.

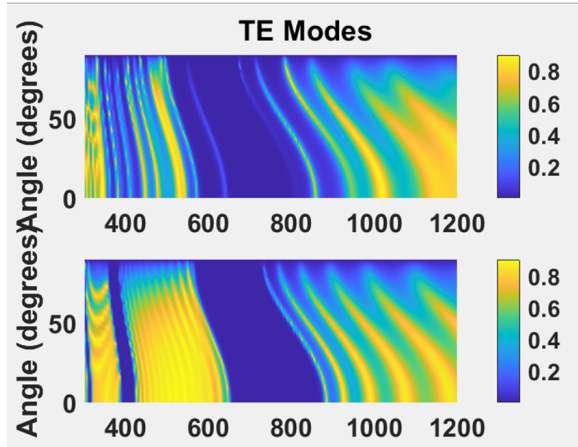


Figure 3: TE modes $\text{SiO}_2/\text{TiO}_2$ for disordered and ordered structures without absorption

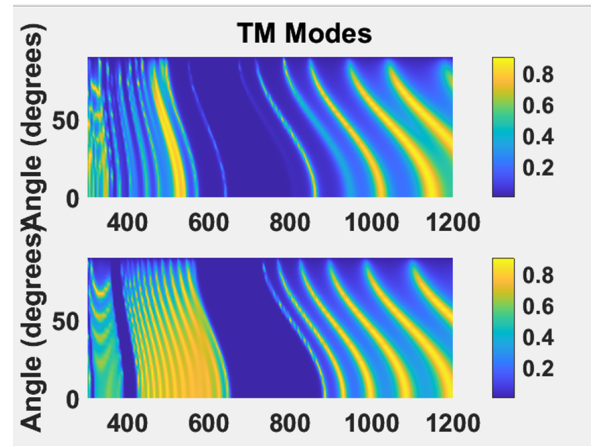


Figure 4: TM modes $\text{SiO}_2/\text{TiO}_2$ for disordered and ordered structures without absorption

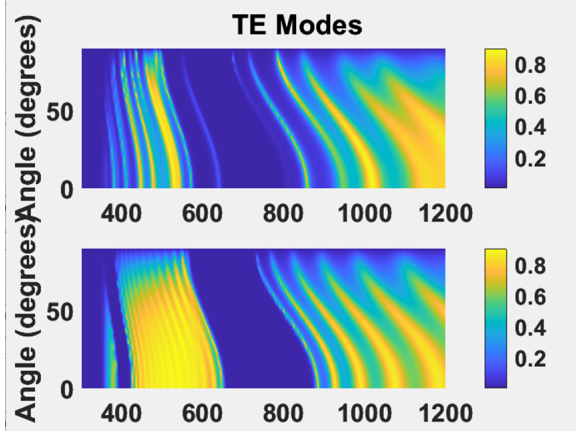


Figure 5: TE modes $\text{SiO}_2/\text{TiO}_2$ for disordered and ordered structures with absorption

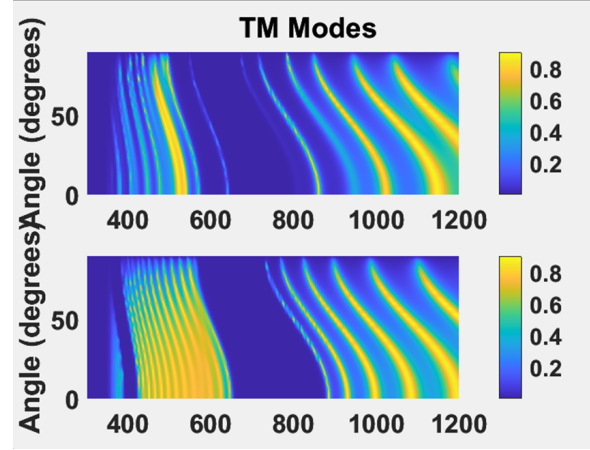


Figure 6: TM modes $\text{SiO}_2/\text{TiO}_2$ for disordered and ordered structures with absorption

5 Additional simulations

5.1 Simulation with GaAs/ SiO_2 multilayers

In this section, I simulate a one-dimensional photonic crystal composed of alternating layers of GaAs and SiO_2 . The structure has a periodic configuration with layer thicknesses $d_{\text{GaAs}} = 60$ nm and $d_{\text{SiO}_2} = 100$ nm. As in the previous cases, I analyze both ordered and disordered configurations.

Simulations were carried out both with and without considering absorption. For the non-absorbing case, only the real part of the refractive index was used, while in the absorbing case the imaginary part $k(\lambda)$ was also included.

The dispersion of GaAs was taken into account using available experimental data reported in [Ref. 6], from which both the real and imaginary parts of the refractive index were extracted and interpolated over the wavelength range of 300 nm to 800 nm, with a step of 5 nm. Beyond 800 nm, the interpolation is no longer valid and simulation results in that region should be disregarded.

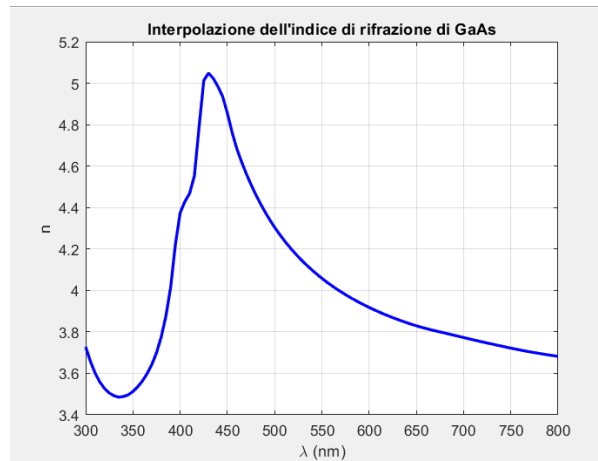


Figure 7: Refractive index of GaAs

Figure 8 and Figure 9 show the transmission spectra of ordered and disordered GaAs/SiO₂ multilayers, respectively, without and with absorption. Without absorption, the periodicity of the GaAs/SiO₂ multilayer allows transmission in the range of 525 nm to 775 nm, with several band gaps appearing between 465 nm and 525 nm, as well as around 420–445 nm and 370–395 nm. On the other hand, the presence of absorption completely reduces the transmission peaks and smooths the spectral features due to light attenuation in GaAs.

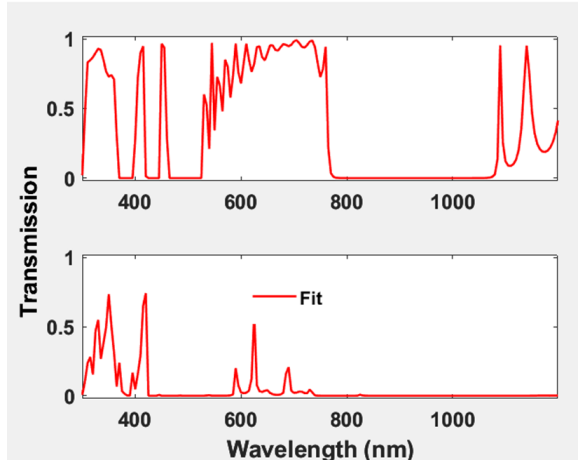


Figure 8: GaAs/SiO₂ photonic crystal without absorption, for both ordered and disordered structures.

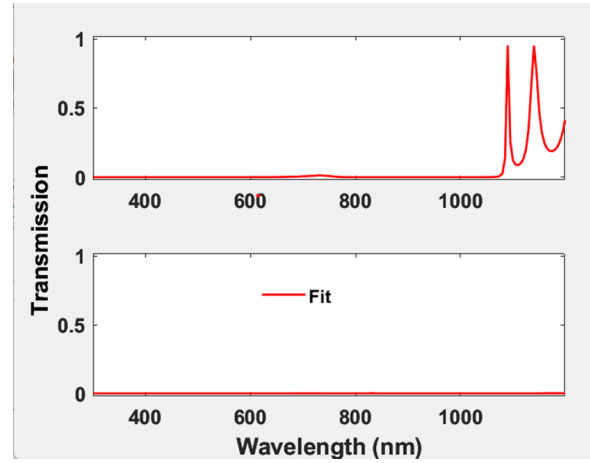


Figure 9: GaAs/SiO₂ photonic crystal with absorption, for both ordered and disordered structures.

To complement the analysis, I also investigated the angular dependence of the structure for both TE and TM modes, in absence of absorption. The results show the expected blue-shift of the photonic bandgap with increasing angle of incidence, consistent with theoretical predictions based on the transfer matrix method.

The case with absorption is not depicted for obvious reasons.

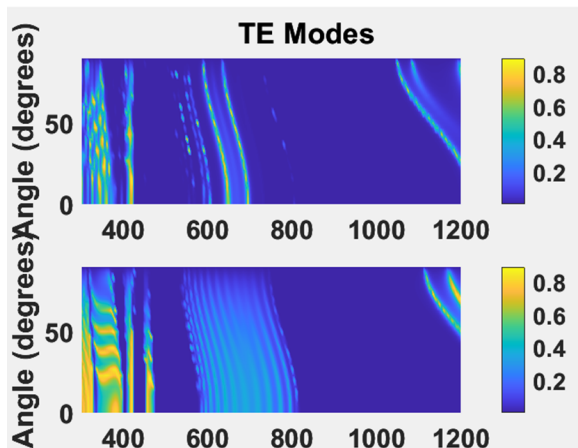


Figure 10: TE mode for disordered and ordered GaAs/SiO₂ structures

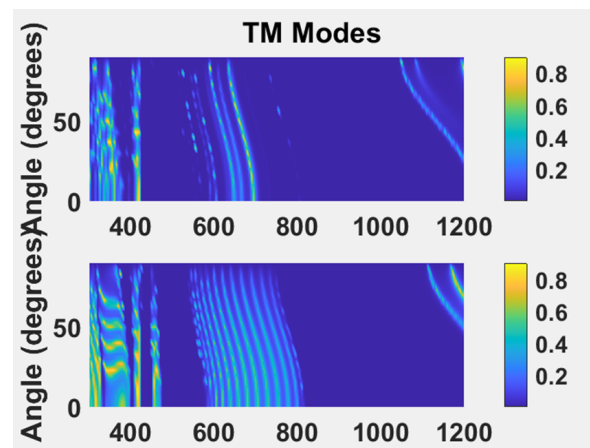


Figure 11: TM mode for disordered and ordered GaAs/SiO₂ structures

5.2 Microcavity creation via defect layer or third material inclusion

In this section, I investigate the formation of an optical cavity in a 1D photonic crystal by introducing a structural defect or a third material. The comparison involves three multilayer configurations:

- **Periodic structure:** A reference structure with periodic layers of SiO_2 and TiO_2 , each of thickness $d = 100$ nm. This configuration gives rise to a well-defined photonic band gap in the range of 655 nm to 870 nm, where light propagation is significantly suppressed.
- **Defect layer cavity:** A modified structure with the same periodicity as above, but including a single TiO_2 defect layer of increased thickness $d = 145$ nm in the center of the multilayer. The introduction of this thicker layer creates a sharp transmission peak at 845 nm, corresponding to a localized cavity mode.
- **Third material (ZrO_2) cavity:** A cavity is created by inserting a layer of ZrO_2 (n_{ZrO_2}). All layers are kept at 100 nm thickness. ZrO_2 is selected because, according to Ref. [7], its extinction coefficient k is negligible in the 300–1200 nm spectral range, making it suitable for low-loss designs. This structure opens a band gap from 630 nm to 915 nm, with a distinct transmission peak centered at 770 nm.

The coefficients of ZrO_2 can be obtained from [Ref. 2]

$$n_{\text{ZrO}_2}(\lambda) = \sqrt{1 + \frac{1.347091 \cdot \lambda^2}{\lambda^2 - 0.062543^2} + \frac{2.117788 \cdot \lambda^2}{\lambda^2 - 0.166739^2} + \frac{9.452943 \cdot \lambda^2}{\lambda^2 - 24.32057^2}} \quad (14)$$

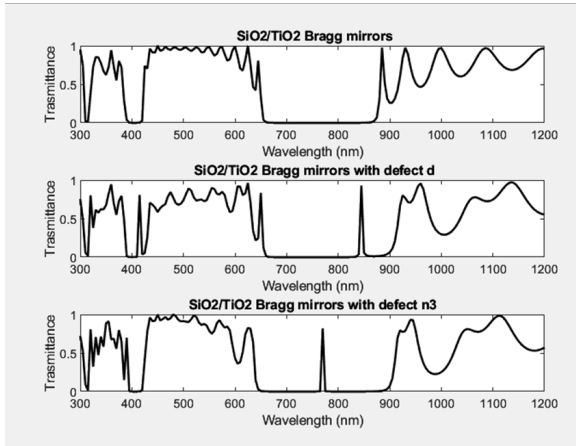


Figure 12: Comparison of Bragg mirror structures: (a) periodic $\text{SiO}_2/\text{TiO}_2$ Bragg mirror, (b) Bragg mirror with a TiO_2 defect, (c) Bragg mirror with ZrO_2 defect.

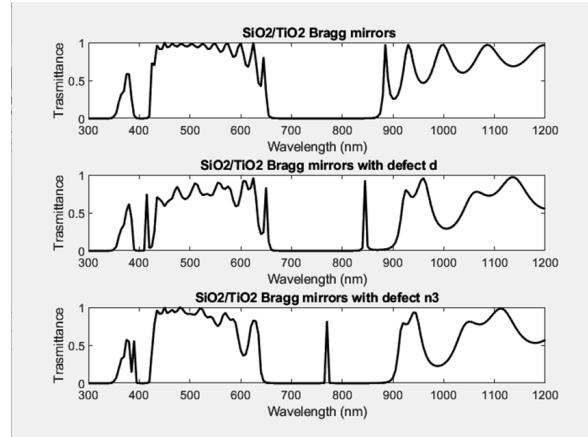


Figure 13: Comparison of Bragg mirror structures: (a) periodic $\text{SiO}_2/\text{TiO}_2$ Bragg mirror, (b) Bragg mirror with a TiO_2 defect, (c) Bragg mirror with ZrO_2 defect, with absorption.

Comment

The creation of a microcavity is particularly useful for the development of laser system, but now also for single-photon sources. In particular, placing a quantum dot — a structure confined in all three spatial dimensions — in the defect layer allows for a suitable spontaneous emission rate. This rate is enhanced by the reduced volume of the cavity, the high quality factor of the cavity's resonant mode, and the frequency matching between the artificial atom (the quantum dot), with transition frequency ω_0 , and the resonant frequency of the cavity.

6 Summary of Main Results

- **Absorption Effects:** Including the imaginary part of the refractive index improves the match between simulations and experiments, especially in the UV region (300–400 nm), where materials like TiO_2 exhibit non-negligible absorption.
- **Angular Dependence:** Increasing the angle of incidence causes a noticeable blue-shift (from 75 up to 150 nm) of the photonic bandgap.
- **Material Substitution:** Replacing TiO_2 with GaAs introduces significant absorption, which suppresses transmission (300–800 nm). Without absorption, GaAs/ SiO_2 structures exhibit multiple bandgaps and sharp transmission features.
- **Microcavity Modes:** Introducing a structural defect (e.g., a thicker TiO_2 layer) or a third material (e.g., ZrO_2) creates localized optical modes. These appear as narrow transmission peaks inside the photonic bandgap, which are useful for designing optical resonators and single-photon sources.

References and Notes

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